

Uniqueness idea and the Uniqueness Theorem for Laplace's equation

Review: why we need boundary/initial data for uniqueness

Consider the ODE

$$\frac{d^2x}{dt^2} + k^2x = 0.$$

Its general solution is

$$x(t) = A \cos(kt) + B \sin(kt),$$

so without additional conditions there are infinitely many solutions (not unique).

To make the solution *unique*, we must specify enough information. For a second-order ODE, two independent conditions are needed (e.g., initial values).

Example Given $x(0) = 0$ and $x\left(\frac{\pi}{2k}\right) = 2$,

$$x(0) = A = 0, \quad x\left(\frac{\pi}{2k}\right) = B \sin\left(k \frac{\pi}{2k}\right) = B \cdot 1 = B = 2.$$

Hence the **unique** solution is

$$x(t) = 2 \sin(kt).$$

Laplace equation: a second-order PDE

In electrostatics (charge-free region),

$$\nabla^2 \phi = 0 \quad (\text{Laplace's equation, a 2nd-order PDE}).$$

Just like the ODE case, a PDE needs *boundary conditions* to make the solution unique.

First Uniqueness Theorem (Dirichlet uniqueness for Laplace)

First Uniqueness Theorem

Let V be a volume with boundary surface $S = \partial V$. If ϕ satisfies Laplace's equation in V ,

$$\nabla^2 \phi = 0 \quad \text{in } V,$$

and if the value of ϕ is specified everywhere on the boundary S (Dirichlet boundary condition),

$$\phi = \phi_S \quad \text{on } S,$$

then the solution ϕ inside V is **uniquely determined**.



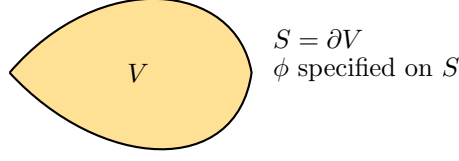


Figure 1: Dirichlet problem: specify ϕ on the boundary S , determine ϕ uniquely inside V .

Proof:

Assume there are two solutions ϕ_1 and ϕ_2 such that

$$\nabla^2 \phi_1 = 0, \quad \nabla^2 \phi_2 = 0 \quad \text{in } V, \quad \phi_1 = \phi_2 = \phi_S \quad \text{on } S.$$

Define their difference:

$$\psi \equiv \phi_1 - \phi_2.$$

Then

$$\nabla^2 \psi = \nabla^2 \phi_1 - \nabla^2 \phi_2 = 0 \quad \text{in } V, \quad \psi = 0 \quad \text{on } S.$$

Now use the identity

$$\nabla \cdot (\psi \nabla \psi) = |\nabla \psi|^2 + \psi \nabla^2 \psi.$$

Integrate over V :

$$\int_V \nabla \cdot (\psi \nabla \psi) dv = \int_V |\nabla \psi|^2 dv + \int_V \psi \nabla^2 \psi dv.$$

But $\nabla^2 \psi = 0$, so

$$\int_V \nabla \cdot (\psi \nabla \psi) dv = \int_V |\nabla \psi|^2 dv.$$

Apply divergence theorem:

$$\oint_S \psi \nabla \psi \cdot d\vec{S} = \int_V |\nabla \psi|^2 dv.$$

Since $\psi = 0$ on S , the surface integral is zero, hence

$$\int_V |\nabla \psi|^2 dv = 0.$$

Because $|\nabla \psi|^2 \geq 0$ everywhere, the only way its integral is zero is

$$\nabla \psi = \vec{0} \quad \text{everywhere in } V \quad \Rightarrow \quad \psi = \text{constant in } V.$$

But $\psi = 0$ on S , so that constant must be zero. Therefore

$$\boxed{\psi = 0 \text{ in } V \Rightarrow \phi_1 = \phi_2 \text{ in } V.}$$

Hence the solution is unique.

Key takeaway

Laplace is a 2nd-order PDE. If ϕ is specified on the boundary S , then ϕ inside is unique. Proof idea: assume two solutions, subtract, show $\int_V |\nabla \psi|^2 dv = 0 \Rightarrow \psi = 0$.

Additional proof idea (as discussed in class): take the difference $\phi_3 = \phi_2 - \phi_1$

Trick: Suppose there are two solutions ϕ_1 and ϕ_2 to the *same* boundary-value problem. Define

$$\phi_3 \triangleq \phi_2 - \phi_1.$$

Case A: Laplace equation (charge-free region). Assume

$$\nabla^2 \phi_1 = 0, \quad \nabla^2 \phi_2 = 0 \quad \text{in } V,$$

and both satisfy the *same* Dirichlet boundary condition

$$\phi_1 = \phi_2 = \phi_S \quad \text{on } S = \partial V.$$

Then, by linearity of Laplace operator,

$$\nabla^2 \phi_3 = \nabla^2(\phi_2 - \phi_1) = \nabla^2 \phi_2 - \nabla^2 \phi_1 = 0.$$

Also, on the boundary,

$$\phi_3|_S = \phi_2 - \phi_1 = \phi_S - \phi_S = 0.$$

Now invoke the **maximum/minimum principle for Laplace equation**:

Maximum/Minimum principle (self-consistent statement)

Key idea: Laplace equation has no interior extrema.

If u satisfies Laplace's equation

$$\nabla^2 u = 0 \quad \text{in a bounded region } V,$$

and u is continuous on the closed region $\bar{V}$, then:

- u cannot have a *local* maximum or a *local* minimum at any point strictly inside V , unless u is a constant everywhere in V .
- Therefore, any maximum or minimum (extreme value) of u in $\bar{V}$ must occur on the boundary $S = \partial V$.

Since $\phi_3 = 0$ on all of S , both the maximum and minimum values of ϕ_3 on the boundary are zero. Therefore, the maximum and minimum of ϕ_3 in the entire region must also be zero, which implies

$$\phi_3 = 0 \text{ everywhere in } V.$$

Hence,

$$\phi_2 = \phi_1 \Rightarrow \text{solution is unique.}$$

Case B: Poisson equation (with charge in the volume). Now assume there is a volume charge density $\rho(\mathbf{r})$ inside V , and both solutions satisfy the *same* Poisson equation:

$$\nabla^2 \phi_1 = -\frac{\rho}{\epsilon_0}, \quad \nabla^2 \phi_2 = -\frac{\rho}{\epsilon_0} \quad \text{in } V,$$

along with the same Dirichlet boundary condition $\phi_1 = \phi_2 = \phi_S$ on S . Then the difference again satisfies Laplace:

$$\nabla^2 \phi_3 = \nabla^2(\phi_2 - \phi_1) = \left(-\frac{\rho}{\epsilon_0}\right) - \left(-\frac{\rho}{\epsilon_0}\right) = 0,$$

and $\phi_3 = 0$ on S . By the same maximum-principle argument,

$$\phi_3 = 0 \text{ in } V \Rightarrow \phi_2 = \phi_1.$$

So the Poisson problem is also unique *provided both the source ρ and the boundary potential are the same.*

Corollary (to the First Uniqueness Theorem)

(Poisson/Dirichlet uniqueness). The electrostatic potential ϕ in a volume V is **uniquely determined** if:

- the charge density $\rho(\mathbf{r})$ is specified everywhere inside V (so $\nabla^2 \phi = -\rho/\epsilon_0$ is known), and
- the potential ϕ is specified on the entire boundary surface $S = \partial V$.

$$\rho \text{ in } V \text{ and } \phi \text{ on } S \implies \phi \text{ is unique in } V.$$

Second Uniqueness Theorem (related to conductors)

In many electrostatic problems, we may *not* know the potential on every boundary surface. Instead, we know the **total free charge** on each conducting surface. This leads to the second uniqueness theorem.

Second Uniqueness Theorem (statement)

Consider a volume V containing a specified volume free-charge density $\rho(\mathbf{r})$, and bounded by conductors. If the **total free charge** on each conductor (and on the outer bounding conductor, if present) is specified, then the **electrostatic field $\mathbf{E}$** in the region between the conductors is **uniquely determined** (up to an arbitrary additive constant in potential).



Note (important precision). Since $\mathbf{E} = -\nabla\phi$, uniqueness of $\mathbf{E}$ implies ϕ is unique up to an additive constant (which is expected unless one conductor potential is fixed as a reference).

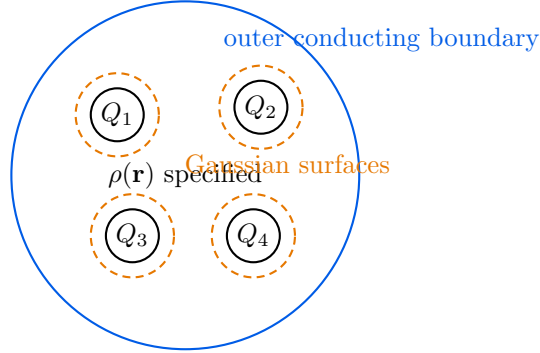


Figure 2: Volume V bounded by conductors: total free charge on each conductor is specified.

Proof :

Assume (for contradiction) that there exist *two* different electrostatic fields $\mathbf{E}_1$ and $\mathbf{E}_2$ in the region between conductors that satisfy the same given data.

(i) **Same volume charge density ρ .** In the region between conductors, Gauss's law (differential form) gives

$$\nabla \cdot \mathbf{E}_1 = \frac{\rho}{\varepsilon_0}, \quad \nabla \cdot \mathbf{E}_2 = \frac{\rho}{\varepsilon_0}.$$

(ii) **Same total charge on each conducting surface.** For each conductor surface S_i , the total free charge is specified as Q_i . Hence Gauss's law (integral form) gives

$$\oint_{S_i} \mathbf{E}_1 \cdot d\mathbf{S} = \frac{Q_i}{\varepsilon_0}, \quad \oint_{S_i} \mathbf{E}_2 \cdot d\mathbf{S} = \frac{Q_i}{\varepsilon_0}.$$

Similarly, for the outer conducting boundary surface S_{out} (if present) with specified total charge Q_{out} ,

$$\oint_{S_{\text{out}}} \mathbf{E}_1 \cdot d\mathbf{S} = \frac{Q_{\text{out}}}{\varepsilon_0}, \quad \oint_{S_{\text{out}}} \mathbf{E}_2 \cdot d\mathbf{S} = \frac{Q_{\text{out}}}{\varepsilon_0}.$$

(iii) **Take the difference field (same trick as before).** Define

$$\mathbf{E}_3 \triangleq \mathbf{E}_1 - \mathbf{E}_2.$$

Then, subtracting the divergence equations,

$$\nabla \cdot \mathbf{E}_3 = \nabla \cdot \mathbf{E}_1 - \nabla \cdot \mathbf{E}_2 = \frac{\rho}{\varepsilon_0} - \frac{\rho}{\varepsilon_0} = 0.$$

Hence,

$$\nabla \cdot \mathbf{E}_3 = 0 \quad \text{in the region between conductors.}$$

Also, subtracting the flux conditions on each conductor (and on the outer boundary),

$$\oint_{S_i} \mathbf{E}_3 \cdot d\mathbf{S} = \oint_{S_i} \mathbf{E}_1 \cdot d\mathbf{S} - \oint_{S_i} \mathbf{E}_2 \cdot d\mathbf{S} = \frac{Q_i}{\varepsilon_0} - \frac{Q_i}{\varepsilon_0} = 0,$$

and similarly $\oint_{S_{\text{out}}} \mathbf{E}_3 \cdot d\mathbf{S} = 0$.

Recall we defined

$$\mathbf{E}_3 = \mathbf{E}_1 - \mathbf{E}_2,$$

and we already obtained:

$$\nabla \cdot \mathbf{E}_3 = 0 \quad \text{in the region between conductors,} \quad \oint_{S_k} \mathbf{E}_3 \cdot d\mathbf{S} = 0 \quad \text{for every conducting boundary } S_k.$$

Step 1: Flux is zero on each conducting boundary (given). For each conducting boundary surface S_k ,

$$\oint_{S_k} \mathbf{E}_3 \cdot d\mathbf{S} = 0.$$

Step 2 (key trick): conductors are equipotential. Even though we do not know how the charge is distributed on each conductor surface, we do know that each conductor surface is an **equipotential**. Let the (constant) potential of conductor k be V_k .

Since $\mathbf{E} = -\nabla\phi$, the potential difference between two conductor surfaces is path-independent. In particular, for $\mathbf{E}_1$ and $\mathbf{E}_2$,

$$-\int_1^2 \mathbf{E}_1 \cdot d\mathbf{l} = V_1 - V_2, \quad -\int_1^2 \mathbf{E}_2 \cdot d\mathbf{l} = V_1 - V_2,$$

(where the endpoints lie on conductor-1 and conductor-2, respectively). Subtracting these two relations gives

$$-\int_1^2 \mathbf{E}_3 \cdot d\mathbf{l} = 0.$$

Hence $\mathbf{E}_3$ is also conservative and we can write

$$\mathbf{E}_3 = -\nabla V_3$$

for some scalar function V_3 . Moreover, because each conducting surface is equipotential, V_3 must be *constant* on each conducting boundary surface.

Step 3: Use the chain rule identity Consider

$$\nabla \cdot (V_3 \mathbf{E}_3) = V_3 (\nabla \cdot \mathbf{E}_3) + \mathbf{E}_3 \cdot (\nabla V_3).$$



But $\nabla \cdot \mathbf{E}_3 = 0$, and $\nabla V_3 = -\mathbf{E}_3$. Therefore,

$$\nabla \cdot (V_3 \mathbf{E}_3) = \mathbf{E}_3 \cdot (-\mathbf{E}_3) = -(\mathbf{E}_3)^2.$$

So we have the key relation

$$\nabla \cdot (V_3 \mathbf{E}_3) = -(\mathbf{E}_3)^2.$$

Step 4: Apply divergence theorem over the region between conductors. Integrate over the volume V (the region between conductors) and apply the divergence theorem:

$$\int_V \nabla \cdot (V_3 \mathbf{E}_3) dv = \oint_S V_3 \mathbf{E}_3 \cdot d\mathbf{S} = - \int_V (\mathbf{E}_3)^2 dv,$$

where S is the union of all conducting boundary surfaces (including the outer conducting boundary, if present).

Step 5: Show the surface integral is zero. On each conductor surface S_k , V_3 is a constant (call it c_k). Hence

$$\oint_{S_k} V_3 \mathbf{E}_3 \cdot d\mathbf{S} = c_k \oint_{S_k} \mathbf{E}_3 \cdot d\mathbf{S} = c_k \cdot 0 = 0,$$

because we already proved $\oint_{S_k} \mathbf{E}_3 \cdot d\mathbf{S} = 0$ for every conductor boundary. Summing over all boundary surfaces,

$$\oint_S V_3 \mathbf{E}_3 \cdot d\mathbf{S} = 0.$$

Therefore the volume integral must satisfy

$$0 = - \int_V (\mathbf{E}_3)^2 dv \Rightarrow \int_V (\mathbf{E}_3)^2 dv = 0.$$

Since $(\mathbf{E}_3)^2 \geq 0$ everywhere, the only way the integral is zero is

$$\mathbf{E}_3 = \mathbf{0} \text{ everywhere in } V.$$

Thus,

$$\mathbf{E}_1 = \mathbf{E}_2.$$

■ (QED)

Lecture conclusion: Uniqueness theorems (main points)

- **First uniqueness theorem (Dirichlet).** If ϕ is specified on the boundary $S = \partial V$ (and ρ is specified for Poisson), then ϕ inside V is unique.
- **Corollary (Poisson/Dirichlet).**

$$\rho(\mathbf{r}) \text{ given in } V \text{ and } \phi \text{ given on } S \Rightarrow \phi \text{ unique in } V.$$

- **Second uniqueness theorem (conductors / charge-specified).** If the region is bounded by conductors, $\rho(\mathbf{r})$ is specified in the volume, and the *total free charge* on each conducting surface is specified, then the electric field $\mathbf{E}$ in the region between conductors is uniquely determined (and ϕ is unique up to an additive constant).
- **Core proof tricks.**
 1. Define the difference field $\mathbf{E}_3 = \mathbf{E}_1 - \mathbf{E}_2$.
 2. Use same data $\Rightarrow \nabla \cdot \mathbf{E}_3 = 0$ and $\oint_{S_k} \mathbf{E}_3 \cdot d\mathbf{S} = 0$ on each conductor.
 3. Use equipotential property $\Rightarrow V_3 = \text{constant}$ on each conductor and $\mathbf{E}_3 = -\nabla V_3$.
 4. Apply $\nabla \cdot (V_3 \mathbf{E}_3) = -(\mathbf{E}_3)^2$ and divergence theorem to force $\mathbf{E}_3 = \mathbf{0}$.

Alternative Proof (more intuitive; avoids chain rule/divergence theorem)

Why this alternate proof is simpler: The previous proof uses a vector-calculus identity and divergence theorem. Here we use an idea we already know from capacitors: *in electrostatics, charges on conductors depend only on potential differences and this dependence is linear.*

We already reached (from the difference-field step)

$$\nabla \cdot \mathbf{E}_3 = 0 \quad \text{in the region between conductors,} \quad \oint_{S_k} \mathbf{E}_3 \cdot d\mathbf{S} = 0 \quad \text{for each conductor surface } S_k,$$

and $\mathbf{E}_3$ is electrostatic, hence conservative:

$$\nabla \times \mathbf{E}_3 = 0 \quad \Rightarrow \quad \mathbf{E}_3 = -\nabla \phi_3.$$

So ϕ_3 satisfies Laplace equation in the region between conductors:

$$\nabla^2 \phi_3 = 0.$$

Also, each conductor is an equipotential, hence on conductor k ,

$$\phi_3 = V_{3k} \quad (\text{a constant on that conductor surface}).$$

So the only remaining question is: *are the constants V_{3k} all the same or can they be different?*

Key physical fact (multi-conductor generalization of $Q = CV$): For a system of conductors, electrostatics is linear, so the vector of total conductor charges $\{Q_k\}$ depends linearly on the conductor potentials $\{V_k\}$. This is commonly written using the **capacitance-coefficient matrix**:

$$\begin{bmatrix} Q_1 \\ Q_2 \\ \vdots \\ Q_N \end{bmatrix} = \mathbf{C} \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_N \end{bmatrix}, \quad (\text{for a fixed geometry, in a charge-free region}).$$

A basic property (also consistent with $Q = C(V_1 - V_2)$ for a two-conductor capacitor) is:



- Charges depend only on *potential differences*. If we add the same constant to all V_k , the charges do not change.
- Therefore, the only way to make $\{Q_k\} = \mathbf{0}$ for all conductors is to have $\{V_k\}$ be the *same constant* on every conductor (no potential differences).

Apply this to the ϕ_3 problem: in the $\mathbf{E}_3$ (difference) problem, the total charge on *each* conductor is zero, i.e.

$$Q_{3k} = 0 \quad \text{for all conductors } k.$$

Hence the conductor potentials for ϕ_3 cannot have any differences; otherwise a nonzero capacitor-like charge would appear. So we must have

$$V_{31} = V_{32} = \cdots = V_{3N} = (\text{same constant}).$$

Therefore ϕ_3 is constant on *all* conductor boundaries. But then, by Laplace uniqueness (Dirichlet),

$$\phi_3 = \text{constant everywhere in the region.}$$

Hence,

$$\boxed{\mathbf{E}_3 = -\nabla\phi_3 = \mathbf{0} \quad \Rightarrow \quad \mathbf{E}_1 = \mathbf{E}_2.}$$

This completes the proof of the second uniqueness theorem.

Takeaway (alternate proof, self-consistent)

Assume two electrostatic solutions $(\mathbf{E}_1, \phi_1)$ and $(\mathbf{E}_2, \phi_2)$ exist in the same geometry with the same volume charge density $\rho(\mathbf{r})$ and the same *total charge* Q_k on every conductor.

Define the difference:

$$\mathbf{E}_3 = \mathbf{E}_1 - \mathbf{E}_2, \quad \phi_3 = \phi_1 - \phi_2.$$

Then the difference problem has $\rho_3 = 0$ in the volume and $Q_{3k} = 0$ on every conductor, hence

$$\nabla \cdot \mathbf{E}_3 = 0, \quad \nabla \times \mathbf{E}_3 = 0 \quad \Rightarrow \quad \mathbf{E}_3 = -\nabla\phi_3, \quad \nabla^2\phi_3 = 0.$$

Because conductors are equipotential, ϕ_3 is a constant on each conductor surface. Now *in a fixed geometry* (with a reference potential chosen, e.g. outer conductor grounded), any *nonzero* potential difference between conductor surfaces would induce a *nonzero* total charge on at least one conductor (the familiar capacitor idea: charge appears only when there is a potential difference). But here $Q_{3k} = 0$ for all k , so there can be *no* potential difference:

$$\phi_3 \text{ takes the same constant value on all conductors.}$$

Therefore ϕ_3 must be constant everywhere in the region, and hence

$$\boxed{\mathbf{E}_3 = \mathbf{0} \quad \Rightarrow \quad \mathbf{E}_1 = \mathbf{E}_2.}$$